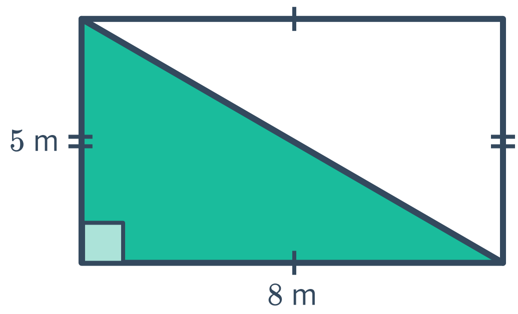
Triangles and rectangles

1 Consider the rectangle shown:

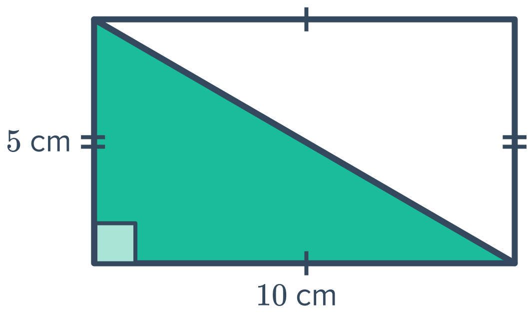
- **a** Find the area of the entire rectangle.
- **b** Find the area of the triangle.
- **c** What do you notice about the two areas?



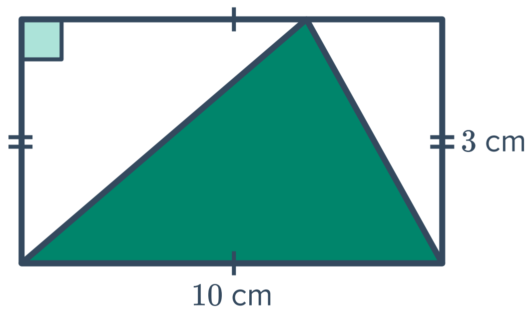
2 For each of the following figures:

- **i** Find the area of the entire rectangle.
- **ii** Find the area of the shaded triangle.

a



b



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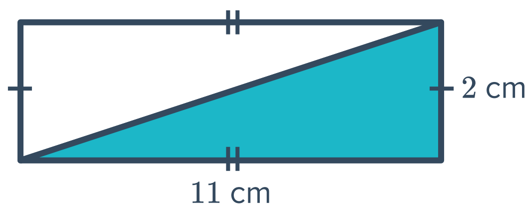

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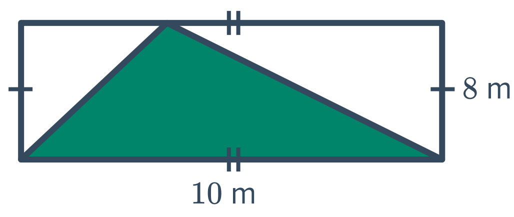

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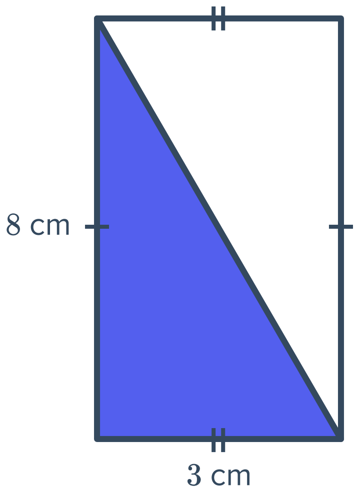

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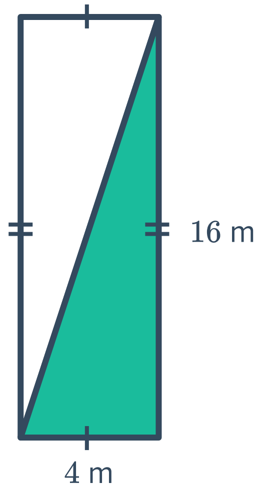
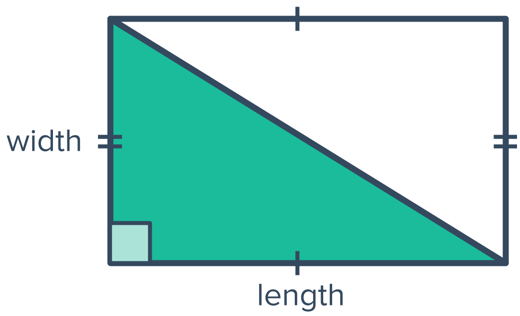


3 Consider the following triangle:

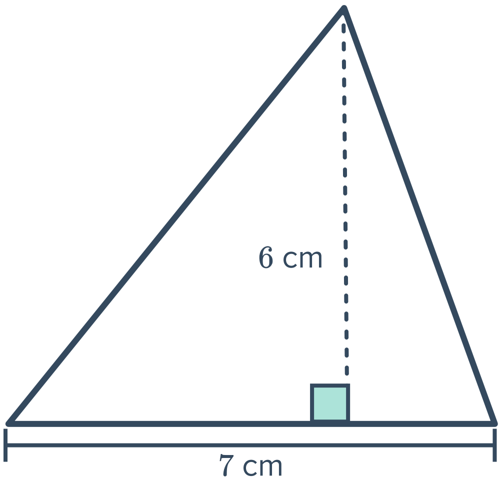
Which dimension of the following rectangle forms the height of the inside triangle?



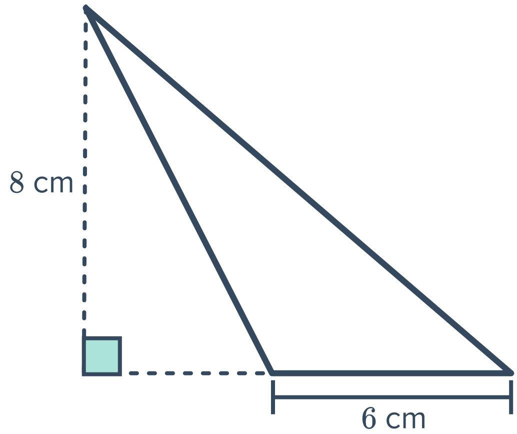
Area of a triangle

4 Find the area of the following triangles:

a



b



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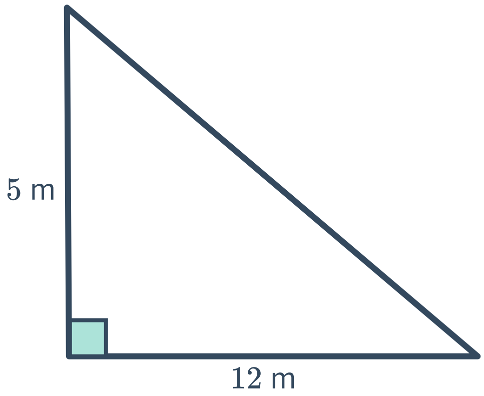

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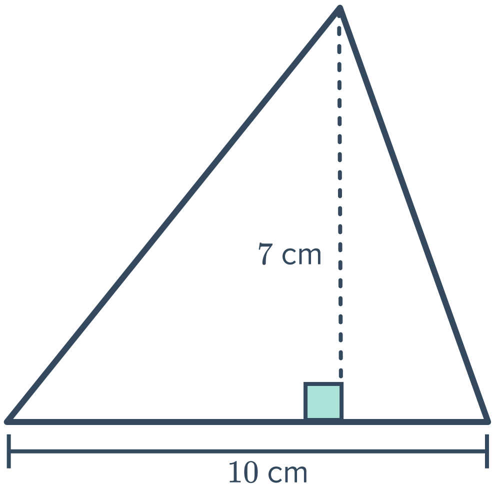

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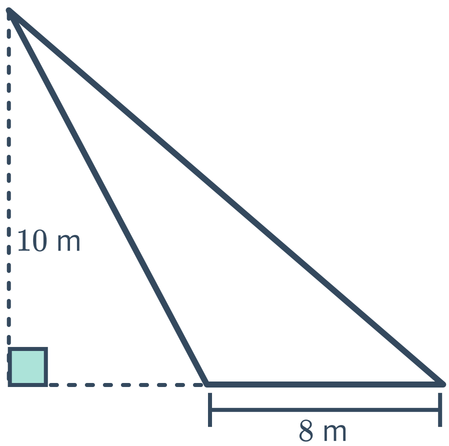


5 Find the area of a triangle which has the f[redacted]ng dimensions:

a Height $= 8$ cm and base $= 9$ cm.

b Height $= 4$ m and base $= 6$ m.

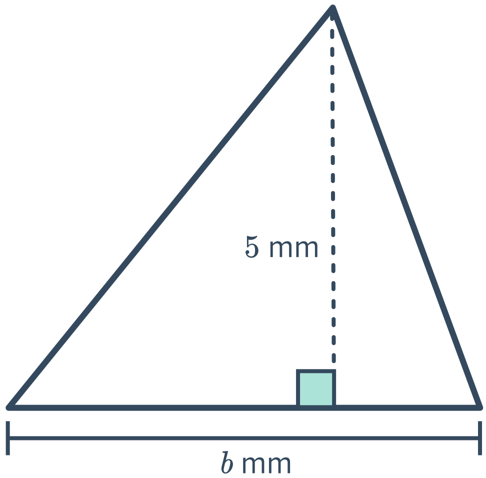
c Height $= 7$ mm and base $= 8$ mm.

d Height $= 3$ cm and base $= 9$ cm.

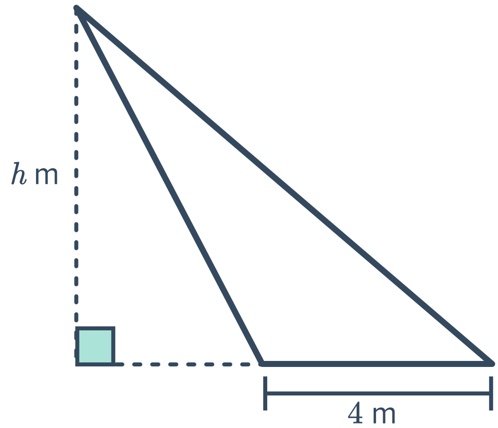
Dimensions from area

6 For each of the following triangles, find the value of b or h given the area:

a Area $= 20 \text{ mm}^2$



b Area $= 48 \text{ m}^2$



7 Determine whether the following could be the dimensions of a triangle with an area of 20 mm^2:

a Base $= 8$ mm, height $= 5$ mm.

b Base $= 1$ mm, height $= 20$ mm.

c Base $= 5$ mm, height $= 8$ mm.

d Base $= 2$ mm, height $= 20$ mm.

8 Complete the following table of base and height measurements for different triangles, which all have an area of 30 m^2:

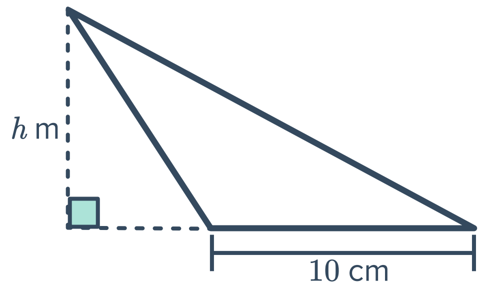
Base (m)		10	15
Height (m)	12		
Area (m^2)	30	30	30

Applications

9 Lisa has purchased a rectangular piece of fabric measuring 6 m in length and 9 m in width.

10 A gutter running along the roof of a house has a cross-section in the shape of a triangle as shown:

If the area of the cross-section is $40\ \mathrm{cm}^2$, and the length of the base of the gutter is $10\ \mathrm{cm}$, find the perpendicular height h of the gutter.



11 The faces on a 4-sided die are all triangular. Each face has a base length of $19\ \mathrm{mm}$ and a perpendicular height of $12\ \mathrm{mm}$. Find the area of one face.